MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 12** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 1: Real Numbers** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, complete prime factorisation or contradiction steps, and boxed final answers with units.
2. In Assertion–Reason questions, truth values of both (A) and (R) as well as whether (R) validly explains (A) are explicitly established.
3. Partial marks are allocated per step in brackets: **[0.5M, 1M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MCQs & Assertion–Reason

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Q1. What is the greatest possible uniform walking speed at which a man can cover distances of 52 km and 91 km in an exact integer number of hours? [1]

(a) 7 km/h (b) 13 km/h (c) 17 km/h (d) 26 km/h

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 13 km/h

Exact integer hours require speed v to divide both 52 and 91. We know:

Formula Used: Greatest uniform speed = HCF$(52, 91)$

Factorising $52 = 2^2 \times 13$ and $91 = 7 \times 13 \implies \text{HCF}(52, 91) = \boxed{13 \text{ km/h}}$. [1M]

Q2. The prime factorisation of 3825 is: [1]

(a) $3 \times 5^2 \times 17$ (b) $3^2 \times 5^2 \times 17$ (c) $3^3 \times 5 \times 17$ (d) $3^2 \times 5^3 \times 17$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) $3^2 \times 5^2 \times 17$

We divide 3825 step by step by primes:

Formula Used: $3825 \div 3 = 1275 \rightarrow 1275 \div 3 = 425 \rightarrow 425 \div 5 = 85 \rightarrow 85 \div 5 = 17$

Thus $3825 = \boxed{3^2 \times 5^2 \times 17}$. [1M]

Q3. The prime factorisation of 5005 is: [1]

(a) $5 \times 7 \times 11 \times 13$ (b) $5^2 \times 7 \times 13$ (c) $5 \times 11^2 \times 13$ (d) $5 \times 7^2 \times 11$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) $5 \times 7 \times 11 \times 13$

Successively dividing 5005 by ascending primes:

Formula Used: $5005 \div 5 = 1001, \quad 1001 = 7 \times 143 = 7 \times 11 \times 13$

Thus $5005 = \boxed{5 \times 7 \times 11 \times 13}$. [1M]

Q4. The prime factorisation of 7429 is: [1]

(a) $13 \times 17 \times 23$ (b) $17 \times 19 \times 23$ (c) $17 \times 19 \times 29$ (d) $19 \times 23 \times 29$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) $17 \times 19 \times 23$

Testing prime divisors of 7429 sequentially:

Formula Used: $7429 \div 17 = 437, \quad 437 \div 19 = 23$

Thus $7429 = \boxed{17 \times 19 \times 23}$. [1M]

Q5. If a and b are two consecutive natural numbers, then HCF(a, b) is: [1]

(a) 0 (b) 1 (c) 2 (d) ab

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 1

Any two consecutive natural numbers $n, n+1$ are co-prime because $(n+1) - n = 1$. We know:

Formula Used: HCF$(n, n+1) = 1$

Therefore HCF$(a, b) = \boxed{1}$. [1M]

Q6. Assertion (A): $\dfrac{2\sqrt{45} + 3\sqrt{20}}{2\sqrt{5}}$ is a rational number.

Reason (R): The square root of every positive integer is an irrational number. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) (A) is true but (R) is false.

Simplifying the radicals in Assertion (A) using $\sqrt{45} = \sqrt{9 \times 5} = 3\sqrt{5}$ and $\sqrt{20} = \sqrt{4 \times 5} = 2\sqrt{5}$:

Formula Used: $\dfrac{2(3\sqrt{5}) + 3(2\sqrt{5})}{2\sqrt{5}} = \dfrac{6\sqrt{5} + 6\sqrt{5}}{2\sqrt{5}} = \dfrac{12\sqrt{5}}{2\sqrt{5}} = 6 \in \mathbb{Q}$

So (A) is **true**. For (R): perfect squares like 4 or 9 give rational square roots ($\sqrt{4} = 2, \sqrt{9} = 3$), so (R) is **false**. Hence option $\boxed{\text{(C)}}$. [1M]

Q7. Find the HCF and LCM of 12, 15, and 21 by applying the prime factorisation method. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Prime factorising each number:

$$12 = 2^2 \times 3^1, \qquad 15 = 3^1 \times 5^1, \qquad 21 = 3^1 \times 7^1. \qquad [1M]$$

We know the prime-exponent HCF and LCM rules:

Formula Used: $\text{HCF} = \prod p^{\min \text{ exp}}, \qquad \text{LCM} = \prod p^{\max \text{ exp}}$

Common prime factor is 3^1:

$$\text{HCF}(12, 15, 21) = \boxed{3}, \qquad \text{LCM}(12, 15, 21) = 2^2 \times 3 \times 5 \times 7 = 4 \times 105 = \boxed{420}. \qquad [1M]$$

Q8. Can two numbers have 15 as their HCF and 175 as their LCM? Give reasons. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

We know the fundamental divisibility condition connecting any two integers' HCF and LCM:

Formula Used: For all $a, b \in \mathbb{N}$, $\text{HCF}(a, b)$ must completely divide $\text{LCM}(a, b)$

Dividing the given LCM 175 by the given HCF 15:

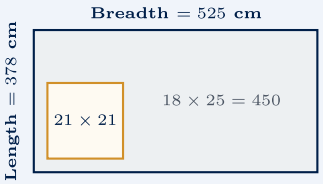
$$175 = 15 \times 11 + 10 \quad (\text{remainder } 10 \neq 0, \text{ or } \frac{175}{15} = \frac{35}{3} \notin \mathbb{Z}). \qquad [1M]$$

Since 15 does not divide 175 evenly, $\boxed{\text{No, two numbers cannot have HCF 15 and LCM 175.}}$ [1M]

SECTION C: Short Answer Type MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. A rectangular courtyard 3.78 m long and 5.25 m broad is to be paved exactly with square tiles of the same size. Find the largest size of such tile and the minimum number of tiles required. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Dimensions in cm: Length $= 3.78$ m $= 378$ cm, Breadth $= 5.25$ m $= 525$ cm.

We know that square tiling without cutting maximizes tile side $s = \text{HCF}(L, B)$:

Formula Used: $s = \text{HCF}(378, 525)$, Min tiles $= \dfrac{378 \times 525}{s^2}$

Prime factorising 378 and 525:

$$378 = 2 \times 27 \times 7 = 2 \times 3^3 \times 7, \qquad 525 = 3 \times 25 \times 7 = 3 \times 5^2 \times 7. \qquad [1M]$$

Common primes 3 and 7 give: $\text{HCF}(378, 525) = 3 \times 7 = \boxed{21 \text{ cm}}$ (size $\boxed{21 \text{ cm} \times 21 \text{ cm}}$). [1M]

Minimum tiles required: $\dfrac{378}{21} \times \dfrac{525}{21} = 18 \times 25 = \boxed{450 \text{ tiles}}$. [1M]

Q10. The traffic lights at three different road crossings change after every 48 seconds, 72 seconds, and 108 seconds respectively. If they change simultaneously at 7:00:00 AM, at what time will they change simultaneously again? [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

The lights next change together after $\text{LCM}(48, 72, 108)$ seconds. We know:

Formula Used: Simultaneous interval $= \text{LCM}(48, 72, 108)$

Prime factorising 48, 72, and 108:

$$48 = 2^4 \times 3, \qquad 72 = 2^3 \times 3^2, \qquad 108 = 2^2 \times 3^3. \qquad [1M]$$

Taking greatest powers of primes 2 and 3:

$$\text{LCM}(48, 72, 108) = 2^4 \times 3^3 = 16 \times 27 = 432 \text{ seconds}. \qquad [1M]$$

Converting 432 seconds into minutes and seconds: $432 = 7 \times 60 + 12 = 7$ min 12 s. Adding to 7:00:00 AM gives $\boxed{7:07:12 \text{ AM}}$. [1M]

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Q11. Find the smallest positive integer which when divided by 16, 20, and 24 leaves remainders 11, 15, and 19 respectively, and is also exactly divisible by 7. [5]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Notice the constant difference between each divisor and its remainder:

$$16 - 11 = 5, \qquad 20 - 15 = 5, \qquad 24 - 19 = 5. \qquad [1M]$$

We know any integer satisfying the remainder conditions has the general form:

Formula Used: $N = k \cdot \text{LCM}(16, 20, 24) - 5 \qquad (k \in \mathbb{N})$

Factorising $16 = 2^4$, $20 = 2^2 \times 5$, $24 = 2^3 \times 3$:

$$\text{LCM}(16, 20, 24) = 2^4 \times 3 \times 5 = 16 \times 15 = 240 \implies N = 240k - 5. \qquad [1.5M]$$

Testing consecutive positive integers $k = 1, 2, 3, \ldots$ for divisibility by 7:

$k = 1 : 240(1) - 5 = 235 = 7 \times 33 + 4$ (no), $k = 2 : 475 = 7 \times 67 + 6$ (no),
$k = 3 : 715 = 7 \times 102 + 1$ (no), $k = 4 : 955 = 7 \times 136 + 3$ (no),
$k = 5 : 1195 = 7 \times 170 + 5$ (no), $k = 6 : 1435 = 7 \times 205 + 0$ (YES!). [2.5M]

Hence the smallest such positive integer is $\boxed{1435}$ $(1435 = 16 \times 89 + 11 = 20 \times 71 + 15 = 24 \times 59 + 19)$.

Q12. Let a and b be two positive integers. (i) If $\text{HCF}(a, b) = 18$ and $\text{LCM}(a, b) = 1080$, and one of the numbers is 216, find the other number. (ii) Can two numbers have 16 as their HCF and 380 as their LCM? Justify mathematically. (iii) If p and q are distinct prime numbers, find $\text{HCF}(p^3q^2, p^2q^4)$ and $\text{LCM}(p^3q^2, p^2q^4)$, and verify that their product equals the product of the two given expressions. [5]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

We know the fundamental HCF–LCM identities:

Formula Used: $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b, \qquad \text{HCF} \mid \text{LCM}$

(i) Finding the other number b: Given $a = 216$, HCF $= 18$, LCM $= 1080$:

$$b = \frac{\text{HCF} \times \text{LCM}}{a} = \frac{18 \times 1080}{216} = \frac{1080}{12} = \boxed{90}. \qquad [1.5M]$$

(ii) Feasibility of HCF $= 16$ and LCM $= 380$: Dividing 380 by 16: $380 = 16 \times 23 + 12$ (remainder $12 \neq 0$). Since HCF must always divide LCM, $\boxed{\text{No, it is impossible.}}$ [1.5M]

(iii) HCF, LCM & product verification for $A = p^3q^2$ and $B = p^2q^4$:

$$\text{HCF}(A, B) = \boxed{p^2q^2}, \qquad \text{LCM}(A, B) = \boxed{p^3q^4}.$$

Verifying products:

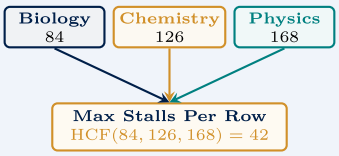
$$\text{HCF} \times \text{LCM} = (p^2q^2)(p^3q^4) = p^5q^6, \qquad A \times B = (p^3q^2)(p^2q^4) = p^5q^6 \implies \text{LHS} = \text{RHS}. \qquad [2M]$$

Q13. CASE STUDY 12: SCIENCE EXHIBITION PODIUM PROJECT DISPLAY LOGISTICS [4 Marks]

A CBSE regional science exhibition is held in an auditorium. Student exhibits are registered in three branches: Biology (84 exhibits), Chemistry (126 exhibits), and Physics (168 exhibits). The organizing committee sets up exhibition aisles such that each row contains the exact same number of project stalls, every row contains exhibits from only one branch of science, and the number of stalls per row is maximized.

(i) Find the maximum number of exhibition stalls in each row. [2]
(ii) Calculate the total number of rows required in the auditorium. [1]
(iii) Find the LCM of 84 and 126 by prime factorisation. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Auditorium Layout: With HCF = 42 project stalls per aisle row, Biology occupies 2 rows, Chemistry 3 rows, and Physics 4 rows, giving 9 uniform rows.

We know the prime-factorisation HCF and LCM rules:

Formula Used: Max stalls/row = HCF(84, 126, 168), LCM(84, 126) = $\prod p^{\max\ \text{exp}}$

(i) Max stalls per row: Factorising 84, 126, and 168:

$$84 = 2^2 \times 3 \times 7, \qquad 126 = 2 \times 3^2 \times 7, \qquad 168 = 2^3 \times 3 \times 7.$$

Taking smallest exponents of common primes 2, 3, and 7:

$$\text{HCF}(84, 126, 168) = 2^1 \times 3^1 \times 7^1 = \boxed{42 \text{ stalls per row}}. \qquad [2M]$$

(ii) Total rows required:

$$\text{Total rows} = \frac{84}{42} + \frac{126}{42} + \frac{168}{42} = 2 + 3 + 4 = \boxed{9 \text{ rows}}. \qquad [1M]$$

(iii) LCM of 84 and 126: Taking highest powers in $84 = 2^2 \times 3 \times 7$ and $126 = 2 \times 3^2 \times 7$:

$$\text{LCM}(84, 126) = 2^2 \times 3^2 \times 7 = 4 \times 9 \times 7 = \boxed{252}. \qquad [1M]$$

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